Wedge connectivity: a family of (n+m)-types over a wedge of an n-connected and an m-connected type has a section

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:wedge-connectivity' (source lines 1600-1614, Lemma 8.6.2); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own definition of the wedge together with the sentence explaining that the statement is reformulated by the induction principle for connected maps, which is exactly what the proof does.

Editorial note. The retained counter values (chapter 8, section 6, theorem offset 1) make the retained lemma print as Lemma 8.6.2, matching the original book numbering.

Editorial note. The first external reference has a label containing a space, which is the author's own identifier 'prop:nconnected_tested_by_lv_n_dependent types'; it is reproduced verbatim. Both external references are navigation pointers only.

Editorial scope. One original result of the HoTT Book, counted once: the wedge connectivity lemma (Lemma 8.6.2).

Allowed standard prerequisites (editorial).

- n-connected pointed types. A pointed type (A,a_0) is n-connected if the map a_0 : unit -> A is n-connected, i.e. if trunc n (fiber of a_0 at a) is contractible for all a.. The hypotheses of the statement are that (A,a_0) is n-connected and (B,b_0) is m-connected, and the proof uses exactly this by viewing a_0 and b_0 as maps out of the unit.
- Induction principle for connected maps. If f : A -> B is n-connected and P : B -> type is a family of n-types, then a section of P o f extends to a section of P.. It is applied twice, once at source line 1619 to the map a_0 : unit -> A and once at source line 1623 to b_0 : unit -> B.
- Fibers and truncation levels of function types. The fiber of precomposition with b_0 over f(a) is the type Sigma k : (prd b:B P(a,b)), f(a) = k(b_0).. It identifies the type Q(a) with that fiber at source lines 1620-1622.

Source reference. Original lemma 7.5.7: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1570–1570. Pointer to the earlier characterization

of n-connected maps by their induction principle relative to families of n-types, which the retained lead-in sentence cites and the retained proof uses in the form stated for type families.

Source reference. Original lemma 8.6.1: sources/hott-homotopy/pinned-original-source.tex, source lines 1569–1569. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

$A \vee B$ is defined to be the pushout of $A \xleftarrow{a_0} \mathbf{1} \xrightarrow{b_0} B$. There is a canonical map $i : A \vee B \to A \times B$ defined by the two maps $\lambda a.\,(a, b_0)$ and $\lambda b.\,(a_0, b)$; the following lemma essentially says that this map is highly connected if A and B are so. It is a bit more convenient both to prove and use, however, if we use the characterization of connectedness from original lemma 7.5.7 and substitute in the universal property of the wedge (generalized to type families).

Lemma 8.6.1 (Wedge connectivity lemma)**.** *Suppose that* (A, a_0) *and* (B, b_0) *are* n*- and* m*-connected pointed types, respectively, with* $n, m \geq 0$*, and let* $P : A \to B \to (n+m)\text{-}\mathsf{Type}$*. Then for any* $f : \prod_{(a:A)} P(a, b_0)$ *and* $g : \prod_{(b:B)} P(a_0, b)$ *with* $p : f(a_0) = g(b_0)$*, there exists* $h : \prod_{(a:A)} \prod_{(b:B)} P(a, b)$ *with homotopies*

$$q : \prod_{a:A} h(a, b_0) = f(a) \qquad \textit{and} \qquad r : \prod_{b:B} h(a_0, b) = g(b)$$

such that $p = q(a_0)^{-1} \cdot r(b_0)$*.*

Proof. Define $Q : A \to \mathcal{U}$ by

$$Q(a) :\equiv \sum_{k:\prod_{(b:B)} P(a,b)} (f(a) = k(b_0)).$$

Then we have $(g, p) : Q(a_0)$. Since $a_0 : \mathbf{1} \to A$ is $(n-1)$-connected, if Q is a family of $(n-1)$-types then we will have $\ell : \prod_{(a:A)} Q(a)$ such that $\ell(a_0) = (g, p)$, in which case we can define $h(a, b) :\equiv \mathsf{pr}_1(\ell(a))(b)$. However, for fixed a, the type $Q(a)$ is the fiber over $f(a)$ of the map

$$\Big(\prod_{b:B} P(a, b)\Big) \to P(a, b_0)$$

given by precomposition with $b_0 : \mathbf{1} \to B$. Since $b_0 : \mathbf{1} \to B$ is $(m-1)$-connected, for this fiber to be $(n-1)$-truncated, by original lemma 8.6.1 it suffices for each type $P(a, b)$ to be an $(n+m)$-type, which we have assumed. □